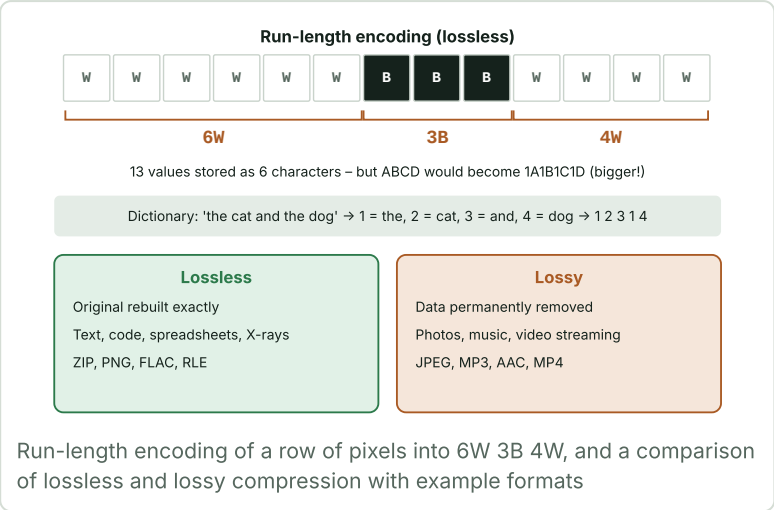
Data compression and file formats

FD13 FACT FILE

Specification: students should be able to

- Explain the need for data compression
- Distinguish between lossy and lossless compression and give appropriate uses of each
- Describe run-length encoding (RLE) and dictionary-based compression and apply them to simple data
- Describe how zipping is used to compress data
- Evaluate common data file formats: txt, wav, bitmap, JPEG, MPEG and GIF
- Calculate uncompressed file sizes and compression ratios

Key words: compression, redundancy, compression ratio, lossless, lossy, run-length encoding (RLE), dictionary-based compression, token, zipping, TXT, WAV, bitmap (BMP), colour depth, resolution (dpi), GIF, JPEG, MPEG (MP4), palette



Run-length encoding of a row of pixels into 6W 3B 4W, and a comparison of lossless and lossy compression with example formats

Why compress data?

Data compression means encoding data using fewer bits than the original representation. It works because most files contain redundancy: the same words, colours or sounds are repeated, and repetition can be stored more efficiently. There are two main reasons for compressing data. First, smaller files take up less storage space on phones, computers and servers. Second, smaller files are quicker to upload, download and stream and use less network bandwidth, which allows web pages to load quickly and video to be watched on slower connections. The cost is processing time to compress and decompress the data, and, with lossy methods, a permanent loss of quality. Images show the need clearly: a six by four inch photograph scanned at 300 dots per inch with 24-bit colour is 1800 by 1200 pixels, which at three bytes per pixel is 6 480 000 bytes, about 6.2 MB. A web page with twenty such images would be very slow to load. The compression ratio, the original size divided by the compressed size, measures how effective compression has been.

Lossy and lossless compression

Lossless compression reduces file size by removing redundancy without discarding any data, so decompression restores the original exactly. It is essential for text, program code, spreadsheets, financial records and medical images, where one wrong character or pixel could matter. Lossy compression permanently removes data that is judged to be less noticeable to people. In JPEG images, fine detail and small colour differences are reduced, because the eye is more sensitive to changes in brightness than in colour. In MP3 audio, frequencies outside the range of human hearing, and quiet sounds masked by louder ones, are removed. In MPEG video, much of each frame is predicted from the frames around it, so only the changes are stored. Lossy methods give much smaller files, often a tenth of the original size or less, and are suitable for photographs, music and video where a perfect copy is not needed. However, the original can never be recovered, higher compression means visibly lower quality, and saving a lossy file repeatedly makes the quality worse each time.

Run-length encoding

Run-length encoding is a simple lossless method that replaces each run of identical values with a count and a single copy of the value. A row of a black-and-white icon containing eight black pixels, sixteen white pixels and eight black pixels can be written as 8B16W8B, seven characters instead of thirty-two, a compression ratio of about 4.6 to 1. To decompress, the program reads each count and writes out the value that many times, restoring the original exactly. RLE works well on images with large areas of a single colour, such as icons, cartoons, logos and scanned documents with white backgrounds. It works badly on data with few repeats, such as photographs or ordinary text, because each single value needs a count as well: ABCD becomes 1A1B1C1D, which is twice as long. In practice, file formats use RLE alongside other techniques or only for runs above a minimum length. When answering questions, show the encoded output, count both sizes carefully and state the ratio or saving.

Dictionary-based compression and zipping

Dictionary-based compression is a lossless method that builds a table of repeated words or sequences and replaces each occurrence in the data with a short token that refers to the table. In the sentence 'the red bus and the blue bus and the green bus', six different words are stored once in a dictionary taking 21 characters, and the sentence becomes eleven one-byte tokens, so 46 bytes become about 32. The saving grows with longer files because common words appear many times while being stored only once. Real compression programs use more advanced versions, such as LZ77 and LZW, which build the dictionary automatically as they read the data, and these are found in ZIP, GIF and PNG. Zipping is the everyday use of this idea: one or more files are compressed losslessly into a single archive with a .zip extension, which is convenient for email attachments, backups and archiving. Any file can be zipped, but formats that are already compressed, such as JPEG or MP4, shrink very little.

Text, sound and image file formats

A plain text file (.txt) stores characters with no formatting, so it can be opened by almost any program on any platform, which makes it useful for exchanging data. A WAV file stores sound as uncompressed digital samples; at CD quality, 44 100 samples per second at 16 bits in stereo, one minute takes about 10.1 MB (10 584 000 bytes), giving excellent quality for recording and editing but large files. MP3 uses lossy compression to reduce this to about 1 MB per minute at 128 kbps. A bitmap (.bmp) divides an image into a grid of pixels and stores each pixel's colour as a binary number; the number of bits per pixel is the colour depth, and 24-bit colour gives 2 to the power 24, or 16 777 216, colours. Bitmaps are uncompressed and very large. When a bitmap is enlarged it becomes pixelated, which is why logos are often drawn as vector graphics instead. File size in bytes is width in pixels multiplied by height in pixels multiplied by colour depth, divided by eight.

GIF, JPEG and MPEG

GIF uses only eight bits per pixel, but each value is an index into a palette of up to 256 colours chosen from the full 24-bit range to suit that particular image. The pixel data is then compressed losslessly. GIF is good for logos, clipart, diagrams and simple animations, where each frame can have its own palette, but it is poor for photographs, which need thousands of shades. JPEG keeps 24-bit colour and uses lossy compression, so it is ideal for photographs with smooth changes of shade; the user can choose the level of compression, trading file size against visible quality, and heavily compressed JPEGs show blocky artefacts. MPEG is a family of standards for compressed video and audio; MPEG-4, usually stored as .mp4, carries both in a single stream with everything needed for playback. Encoding MPEG video takes much more processing than decoding it, so it suits content that is created once and watched many times, such as television catch-up services and online video sites. Choosing a format means balancing quality, file size, compatibility and purpose.

Key terms from the mark schemes

Data compression
Encoding data using fewer bits than the original representation, to reduce storage space and transmission time.

Lossless compression
Compression that reduces file size without losing any data, so the original can be reconstructed exactly.

Lossy compression
Compression that permanently removes data judged less important, giving smaller files that cannot be restored exactly.

Run-length encoding (RLE)
A lossless compression method that replaces a run of identical values with a single value and a count.

Dictionary-based compression
A lossless compression method that stores repeated patterns once in a dictionary and replaces each occurrence with a short reference (token).

Zipping
A lossless method of compressing one or more files into a single archive file with a .zip extension.

Bitmap (BMP)
An uncompressed image format that stores the colour of every pixel in a grid as a binary number.

JPEG
An image format that uses lossy compression on 24-bit colour images, suited to photographs.
Credited: Joint Photographic Experts Group

Compression ratio
The original file size divided by the compressed file size.

Colour depth
The number of bits used to represent the colour of each pixel; n bits give 2^n possible colours, and more bits give a larger file.
Credited: Bit depth

GIF
An image format using up to 256 colours from a palette chosen for the image, compressed losslessly; supports animation.
Credited: Graphics Interchange Format

MPEG
A family of standards for storing compressed video and audio, using lossy compression; MPEG-4 is usually stored as .mp4.
Credited: Moving Picture Experts Group

C C# example

Run-length encoding (RLE)

```
string data = "WWWWWWBBBWWWW";
string result = "";
int i = 0;
while (i < data.Length)
{
    int count = 1;
    while (i + count < data.Length && data[i + count] == data[i]) count++;
    result = result + count + data[i];
    i = i + count;
}
Console.WriteLine(result);   // 6W3B4W
```

Exam focus

Short questions ask for the two reasons for compression: reducing storage space and reducing transmission time. Be precise about the methods: lossy compression permanently removes data, while lossless compression allows the original to be reconstructed exactly. When asked what difference lossless would have made, say there would be no loss of quality but a larger file. Calculation questions usually ask for a bitmap file size; show the pixels as width in inches times dpi multiplied by height in inches times dpi, multiply by the bytes per pixel and state the units, remembering that 24-bit colour is three bytes and gives 16 777 216 colours. Increasing the colour depth increases the file size proportionally. For RLE or dictionary questions show the encoded output and both sizes. When asked why zipping is better than lossy compression for sending a logo to a printer, explain that zipping is lossless, so the printed logo keeps full quality. For file formats, link each one to suitable content: GIF for few colours, JPEG for photographs. A common error is calling RLE or GIF lossy.

? Check yourself

1. State two reasons for compressing files before uploading them to a school website. [2]
2. Explain the difference between lossy and lossless compression. [2]
3. A bitmap image measures 3 inches by 4 inches, has a resolution of 150 dpi and a colour depth of 24 bits. Calculate the file size in bytes. Show your working. [3]

Answers: 1. To reduce the file size so it takes less storage space on the web server (1) / to reduce upload/download (transmission) time so pages load faster (1) 2. Lossy permanently removes some data, so the original cannot be restored exactly (1) / lossless removes redundancy without losing any data, so the original can be reconstructed exactly (1) 3. Pixels: (3 × 150) × (4 × 150) = 450 × 600 = 270 000 pixels (1) / 24 bits = 3 bytes per pixel (1) / 270 000 × 3 = 810 000 bytes (≈ 791 KB) (1)